



7.1 ANALOG INTEGRATED CIRCUITS

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While on the topic of analog circuits, we will look at the LM386 IC. It has all the components for a complete audio-amplifier. Figure 7.3a shows an example of an amplifier made with this integrated circuit, which can be used as a complete amplifier for a walkman, interphone, cassette player or some other audio device. It can also be used as a test circuit for troubleshooting.

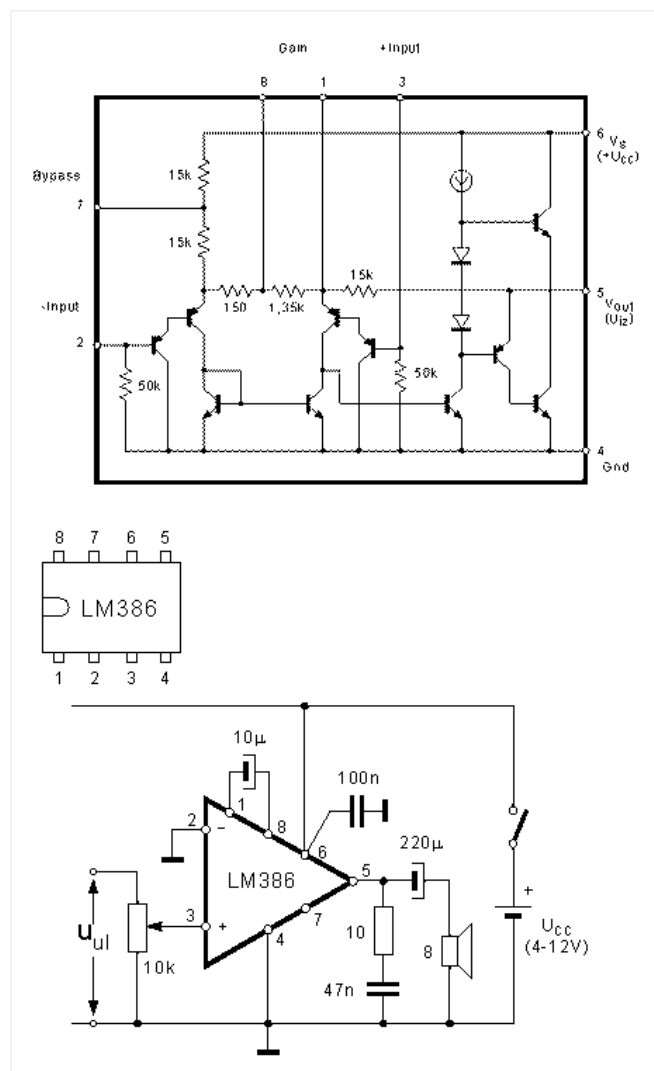


Fig. 7.3: a – A low frequency amplifier using the LM386

The signal is brought to the non-inverting input (between pin 3 and ground). Inverting input (pin 2) is connected to ground. If 10µF is placed between pins 1 and 8 a voltage amplification of 200 is created. If this capacitor is removed the amplification is 20. It is possible to achieve in-between amplification by adding a resistor and

connecting it in series with the capacitor.

One of the essential components in this circuit is the 100nF capacitor which is placed between pin 6 (which is connected to the positive of the supply) and ground. The capacitor should be ceramic and should be mounted as close to the integrated circuit as possible. This is common practice when working with integrated circuits, even when it isn't shown in the diagram as a capacitor connected between the positive and negative stabilizes the voltage and protects the circuit from spikes and a phenomenon called instability. This is due to inductance in the power supply tracks allowing high currents taken by the IC to upset its operation.



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